3D printed house
A Belgian company called Kamp C has 3D printed a complete two storeyed house with a fixed printer. **http://dgit.in/sep20-7**

Face shields for docs
The 3D printing community in Chennai is creating face shields for healthcare professionals. **http://dgit.in/sep20-8**

same year, he started his company called 3D Systems. The company went on to release their first commercial product, the SLA-1, a stereolithographic apparatus machine, in 1988.

Credit: **Machine Design**

The SLA-1, the world's first rapid prototyping printer by 3D Systems

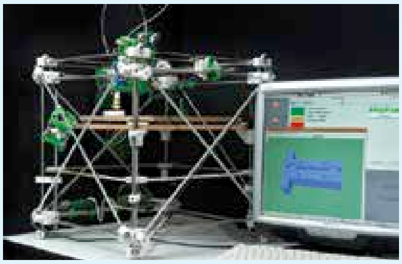
Just a year before the SLA-1 was brought to the public, Carl Deckard at the University of Texas pioneered an alternative method of additive manufacturing called SLS (Selective Laser Sintering), for which he filed for a patent. It used a laser to turn loose powder into a solid, instead of Chuck Hall's liquid resin process. The patent was issued in 1989 to DTM, Inc, which was Deckard's company. The company built the first SLS 3D printer named Betsy and DTM was actually acquired by Hall's 3D Systems soon after.

In 1989, S. Scott Crump and his wife and fellow inventor, Lisa Crump invented and patented a new 3D printing method called Fused Deposition Modelling. The process involves melting a polymer filament and then depositing it onto a substrate, layer by layer, to finally create a three-dimensional object. Crump later went on to co-found Stratasys, which is a prominent 3D printing company even today. Now, these technologies, at the time, were still in their infancy and had some glaring imperfections such as warping in the material as it hardened and the exorbitant cost. However, the potential was undeniable.

THE ERA OF RAPID INNOVATION IN 3D PRINTING

In 1999, the first 3D-printed organ was implanted in humans. Scientists at Wake Forest Institute of Regenerative Medicine printed synthetic biodegradable scaffolds of a human bladder and then layered it with the human patient's cells. The implant had little to no chances of rejection since they were made out of their own cells. Soon after, scientists from different institutions began fabricating 3D printed organs and prosthetic body parts such as a functional mini kidney, a prosthetic leg, and even blood vessels.

The 2000s was the decade when 3D printing merged with the open-source movement. In 2005, Dr Adrian Bowyer launched the RepRap Project, an open-source project aimed to build a 3D printer that could print most of its own parts. The idea behind this was to democratise the 3D printing technology by making it more accessible for everyone. In 2007, Darwin was revealed, which was the first RepRap 3D Printer that could replicate itself by building its own parts.



Darwin, the world's first RepRap 3D Printer

Shapeways, a 3D printing service in the Netherlands released in 2008, that allowed users to submit their own designs for 3D printing. The company 3D printed their designs and shipped it back to them, allowing non-tech population such as artists, architects, and others to make use of the technology. In 2009, Kickstarter launched and the crowdfunding site quickly saw an influx of 3D printers, some of which went on to become major players in the printing industry. The most funded Kickstarter 3D printer project was Micro which raised a staggering $3,401,361 on an ask of $50,000.

In 2009, upon the expiration of the FDM patent held by Stratasys, a company called Makerbot launched which helped to bring 3D printing to the masses. Building upon the success of RepRap, MakerBot started making open-source DIY kits for people wanting to own 3D printers or make 3D printed products. The company created Thingiverse, an online library where users could submit and download 3D printable files. This opened up the doors to collaborations and Thingiverse became the world's largest online 3D printing community. Side note: Stratasys went on to acquire Makerbot for around $400 million in 2013.

3D printing soon after went large-scale and subverted opinions that 3D printing was only suitable for small objects. In 2011, engineers at the University of Southampton designed the world's first unmanned 3D printed aircraft for less than $7,000. In the same year, Urbee, a prototype car with a 3D printed body was presented by Kor Ecologic at the TEDx Winnipeg conference in Canada. Now, 3D printing is even being used to manufacture affordable housing for the developing nations.

The next big advancement in the mass adoption of 3D printing by researchers and scientists was in 2015, when Cellink, a Swedish company, introduced the first standardised commercial bio-ink to the market, priced at $99 per bio-ink cartridge. The bio-ink is derived from seaweed material called non-cellulose alginate and it can be used for printing tissue cartilage. Later that year, the company also introduced the INKREDIBLE 3D printer for $4,999. These products made 3D printing much more affordable allowing widespread adoption by researchers, scientists, and other professionals.

Now, in 2020, with the expiration of most patents and a plethora of open-source projects, 3D printing has seen a massive boom in popularity. There are over 170 3D printer system manufacturers around the world and the number is only increasing every day. The technology is spreading to diverse facets of society such as manufacturing, medicine, education, aerospace, jewelry, and others, since there are more 3D printers available now than ever.